Recall Bias Toward Highly-Cited and English Language Papers in AI Literature Curation

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Abstract

This paper tests whether literature-retrieval pipelines under-retrieve relevant papers that are non-English or have few citations. A synthetic corpus with known topical relevance, language, age, and citation counts is used to compare six pipelines over 20 simulated corpora. In the default setting, English-query BM25 achieves aggregate Recall@100 of 0.836 and a language-recall ratio (LRR) of 0.476; adding a citation boost reduces aggregate recall to 0.704. Multilingual query expansion raises aggregate recall to 0.908 and LRR to 0.919. A diversity-aware re-ranker raises LRR to 0.953 with a small aggregate-recall decrease. In a translation-accuracy sweep, non-English recall crosses the no-translation baseline near $p = 0.25$; this is not the aggregate-recall break-even point, which should be checked against the saved sweep outputs before a precise threshold is claimed. A citation-neutral ranker's low-citation recall gap is confounded by language and text quality, so a popularity-recall ratio below one does not by itself establish a direct popularity effect. The simulated feedback loop adds little change at the default setting. These findings describe the simulator, not real scholarly search systems, and require evaluation on an annotated multilingual corpus before generalization.

Keywords: literature retrieval; group-conditional recall; multilingual information retrieval; citation disparities; ranking fairness; synthetic evaluation; AI-assisted coding.

1 Introduction

Scholarly discovery depends partly on how search systems index, retrieve, and rank papers. English-dominant corpora such as S2ORC [6] and citation-informed representations such as SPECTER [3] are useful resources, but the literature has also documented language barriers in global science [1]. These observations motivate a practical question: can a retrieval pipeline achieve strong aggregate recall while missing relevant papers from particular groups?

In real scholarly search, subgroup failures are difficult to quantify because the complete set of relevant documents is usually unknown and language, age, text quality, and citation counts are entangled. This paper therefore uses a synthetic corpus where topical relevance is known and selected confounders can be varied. It is an initial experiment motivated by a prior course review of recall bias in AI-literature curation [13]; it does not estimate the prevalence or magnitude of bias in a real search engine.

1.1 Research questions

RQ1: Does mixing citation popularity into the ranking score lower recall for low-citation relevant papers and aggregate recall?

RQ2: How large is the recall gap for non-English papers under English-only querying, and how does it depend on cross-lingual overlap?

RQ3: Does multilingual query expansion close the language gap, and how sensitive is it to translation accuracy?

RQ4: Does diversity-aware re-ranking reduce the remaining gap without lowering aggregate recall, and does it risk over-correction?

RQ5: In a citation-neutral ranker, which attributes explain a low-citation recall gap? Since pipeline A never reads citations, can its gap be explained by language and text-quality confounding rather than direct popularity ranking?

RQ6: Does a citation feedback loop measurably amplify disparities over repeated use?

1.2 Contributions

The paper contributes (1) a controlled synthetic testbed with known relevance and tunable language–citation–quality confounding; (2) six retrieval pipelines and group-conditional, age-controlled, and intersectional recall metrics; (3) simulator-specific results for RQ1–RQ6, including low-translation-accuracy and feedback-loop cases; and (4) a transparent account of AI-assisted coding and verification limitations.

2 Related Work and Background

Retrieval and scholarly corpora. BM25 remains a widely used lexical retrieval baseline [7]. S2ORC provides a large corpus of scientific documents [6], while SPECTER uses citation-informed transformer representations [3]. Such resources support scholarly search, but neither corpus scale nor a strong aggregate metric guarantees equal recall across language or citation groups.

Language and multilingual retrieval. Language can be a barrier to scientific communication and discovery [1]. Multilingual benchmarks such as Mr. TyDi [11] and cross-lingual representations such as XLM-R [12] provide useful context for evaluating multilingual retrieval. The present work uses a simpler synthetic translation mechanism rather than a real multilingual model.

Diversity and ranking fairness. Maximal marginal relevance (MMR) is a classic approach to balancing relevance and diversity in ranked results [2]. Fairness-of-exposure work studies how ranking positions distribute visibility across groups [9]; research on scholarly recommender systems also discusses biases and mitigation [5]. This paper focuses specifically on recall conditional on language, age, citation group, and their intersections.

3 Metrics and Definitions

Let $\mathcal{D}_g$ be the set of documents in group g, and let $\mathcal{R}_g$ be the relevant documents in that group across the query set. Group-conditional recall at depth k is

$$\text{Recall}_g @k = \frac{TP_g@k}{|\mathcal{R}_g|}, \tag{1}$$

where $TP_g@k$ is the number of relevant documents from group g retrieved within the top k results for their corresponding queries. In the simulator, a document is relevant to queries for its own topic and irrelevant to queries for other topics.

The language-recall ratio and popularity-recall ratio are

$$\mathrm{LRR} = \frac{\mathrm{Recall}_{\text{non-English}}\ @k}{\mathrm{Recall}_{\text{English}}\ @k}, \qquad \mathrm{PRR} = \frac{\mathrm{Recall}_{\text{low-citation}}\ @k}{\mathrm{Recall}_{\text{high-citation}}\ @k}. \tag{2}$$

A ratio below one indicates lower recall for the numerator group; a ratio above one indicates higher recall. The age-controlled popularity-recall ratio, $\mathrm{PRR}_{\text{age}}$, compares low- and high-citation groups within age bins and aggregates the within-bin comparisons. The intersectional analysis uses eight cells formed by language (English/non-English), recency (recent/old), and citation level (below/above the global median). These ratios are descriptive and do not establish causality without controls for confounding.

4 Methodology

4.1 Why a synthetic testbed?

The experiments were designed in an environment without access to live scholarly indexes. Rather than present synthetic data as real observations, the study uses an explicit generative process. The advantage is that the full relevance denominator is known and confounders can be switched or varied. The limitation is that all numerical results are properties of this simulator, not estimates for Google Scholar, OpenAlex, or AI research assistants.

4.2 Synthetic corpus

Each corpus contains $N = 6{,}000$ documents across $K = 60$ topics, with about 100 relevant documents per topic. Each document has a language (25% non-English), an age in $\{0, \ldots, 15\}$ years, a topical-purity value $u_d \sim \mathrm{Beta}(2, 2)$, and 60 tokens. A fraction $0.08 + 0.40u_d$ of the tokens are topical. Of these topical tokens, 78% are language-specific words from the document's topic, 12% are language-independent tokens, and 10% are noise words from another topic. The English and non-English vocabularies are disjoint apart from shared tokens.

Citation counts are sampled from a Poisson distribution:

$$c_d \sim \mathrm{Poisson}(\lambda_d), \qquad \lambda_d = \exp(0.3 + \log(1 + a_d) + 0.8z_d - \delta\, \mathbb{I}[\text{non-English}] + 0.5\epsilon_d)\,, \tag{3}$$

where a_d is document age, $\mathbb{I}[\cdot]$ is an indicator, and

$$z_d = \rho\widetilde{u}_d + \sqrt{1 - \rho^2}\,\eta_d. \tag{4}$$

Here $\widetilde{u}_d$ is standardised topical purity, and $\eta_d, \epsilon_d \sim \mathcal{N}(0, 1)$. The default language-visibility penalty is $\delta = 0.8$, and the quality–purity correlation parameter is $\rho = 0.3$. Under this construction, non-English papers receive fewer expected citations, and citation counts correlate with age and latent quality. Citation counts do not contain relevance information beyond the text in this simulator.

4.3 Queries and retrieval pipelines

For each topic and each of two repetitions, an English query contains four topical words sampled using Zipf weights and one language-independent token. Thus, each corpus has $60 \times 2 = 120$ queries. For multilingual pipelines, each of the four topical words is expanded with a non-English translation. Each translation is correct with probability p; otherwise, it is replaced by a random word from another topic. The default is $p = 0.8$.

Six pipelines are compared:

A. English BM25: BM25 with $k_1 = 1.2$ and $b = 0.75$, using English queries and separate language-specific collection statistics before scores are merged [7].

B. English BM25 + citation boost: pipeline A with citation score blended in at weight $\beta = 0.3$.

C. English LSA: TF–IDF followed by an 80-dimensional truncated SVD, cosine similarity, and English queries; this is a lightweight semantic-retrieval proxy [4].

D. Multilingual BM25: pipeline A using the expanded multilingual query, without citation information.

E. Multilingual BM25 + diversity re-ranking: pipeline D followed by a greedy MMR-style re-ranker over the top 600 candidates. The score subtracts $\lambda n_c/(s+1)$, where n_c is the number already selected from the same language–recency–citation cell, s is the number selected so far, and $\lambda = 0.25$ [2].

F. Multilingual BM25 + citation boost: pipeline D with citation weight $\beta = 0.3$.

The diversity re-ranker uses the same attributes that define the evaluation groups. Therefore, its parity improvement is partly aligned with the metric by construction and should not be interpreted as a neutral, general-purpose fairness result.

4.4 Protocol and statistics

The main comparison uses 20 independent corpora with seeds 1000–1019; every pipeline is evaluated on identical corpora and queries. The primary metric is Recall@100. Sweeps and feedback runs use 10 corpora per setting. Reported confidence intervals describe variation under the simulator, not real-world uncertainty. At shallow depths, recall is constrained by the number of relevant documents in each group, so parity at very small k is not necessarily a reasonable target.

Table 1: Default simulator and pipeline parameters.

Parameter	Value	Parameter	Value
Documents N	6,000	Language penalty δ	0.8
Topics K	60	Quality–purity correlation ρ	0.3
Tokens per document	60	Translation accuracy p	0.8
Non-English share	0.25	Citation weight β	0.3
Document ages	0–15 years	Diversity strength λ	0.25
Shared-token share	0.12	Re-ranking candidate pool	600
Queries per topic	2	Main depth k	100
Corpora (main/sweeps/feedback)	20/10/10	BM25 (k_1, b)	(1.2, 0.75)

5 Results

5.1 Main comparison (RQ1–RQ4)

Table 2 summarises the six pipelines. The values below are the results reported in the supplied paper; the experiment scripts and JSON outputs have not been independently rerun as part of this document reconstruction.

Table 2: Reported Recall@100 and group-conditional ratios (mean and 95% CI where available, across 20 corpora). English and non-English columns are descriptive means without confidence intervals.

ID	Pipeline	Aggregate	English	Non-English	LRR	PRR_{age}	Worst cell
A	English BM25	0.836 ± 0.002	0.962	0.458	0.476 ± 0.007	0.788 ± 0.006	0.424 ± 0.020
B	English BM25 + citation	0.704 ± 0.006	0.900	0.113	0.126 ± 0.008	0.498 ± 0.009	0.016 ± 0.004
C	English LSA	0.880 ± 0.002	0.999	0.520	0.521 ± 0.006	0.811 ± 0.006	0.464 ± 0.024
D	Multilingual BM25	0.908 ± 0.002	0.927	0.852	0.919 ± 0.006	0.926 ± 0.004	0.830 ± 0.008
E	Multilingual + diversity	0.904 ± 0.002	0.914	0.871	0.953 ± 0.005	0.949 ± 0.004	0.853 ± 0.006
F	Multilingual + citation	0.837 ± 0.005	0.875	0.722	0.825 ± 0.009	0.720 ± 0.010	0.624 ± 0.016

RQ1: Citation popularity lowers recall in this simulator. Adding the citation boost to English BM25 (A→B) reduces aggregate recall from 0.836 to 0.704, a paired difference of −0.132, and reduces PRR_{age} from 0.788 to 0.498. The lowest citation quartile's recall falls from approximately 0.75 to 0.47, while the highest quartile changes little (approximately 0.92 to 0.93). The same pattern appears for multilingual retrieval (D→F): aggregate recall falls by 0.071 and PRR_{age} by 0.206. Because citation counts add no relevance signal beyond text in this simulator, this result does not show that citations are harmful in real search.

RQ2: English-only querying leaves many non-English papers unretrieved. Pipeline A retrieves 96.2% of English relevant papers and 45.8% of non-English relevant papers (LRR = 0.476). The citation-boosted English pipeline is worse for non-English papers, with recall 0.113 and LRR = 0.126. LSA raises non-English recall to 0.520 but does not close the gap.

RQ3: Multilingual query expansion substantially reduces the language gap. Pipeline D reaches aggregate recall 0.908 and LRR = 0.919. Its advantage depends on translation quality: incorrect translations can add misleading terms, and at low translation accuracy multilingual expansion can be worse than using no translation. In the reported sweep, non-English recall crosses the no-translation baseline near $p = 0.25$; this threshold should not be confused with aggregate-recall break-even.

RQ4: Diversity re-ranking gives a small additional parity improvement. Pipeline E raises LRR from 0.919 to 0.953 and PRR_{age} from 0.926 to 0.949, while the worst intersectional cell rises from 0.830 to 0.853. Aggregate recall decreases by 0.005. At perfect translation ($p = 1$), the reported LRR exceeds one (1.027), illustrating possible over-correction. Because the re-ranker explicitly uses the evaluation groups, this result is best treated as a test of a targeted intervention, not independent evidence that generic diversity re-ranking guarantees fairness.

Table 3: Reported paired contrasts across 20 corpora. Differences are first pipeline minus second pipeline.

Contrast	Metric	Difference (95% CI)	p
B vs. A	Aggregate recall	$-0.132\ [-0.137, -0.128]$	< 0.001
B vs. A	PRR_{age}	$-0.290\ [-0.296, -0.283]$	< 0.001
D vs. A	LRR	$+0.443\ [+0.436, +0.450]$	< 0.001
D vs. A	Aggregate recall	$+0.073\ [+0.071, +0.074]$	< 0.001
E vs. D	Aggregate recall	$-0.005\ [-0.005, -0.004]$	< 0.001
E vs. D	LRR	$+0.034\ [+0.032, +0.037]$	< 0.001
E vs. D	PRR_{age}	$+0.022\ [+0.020, +0.025]$	< 0.001
F vs. D	PRR_{age}	$-0.206\ [-0.214, -0.199]$	< 0.001
F vs. D	Aggregate recall	$-0.071\ [-0.075, -0.068]$	< 0.001
D vs. C	Aggregate recall	$+0.029\ [+0.027, +0.031]$	< 0.001

5.2 Intersectional recall

A single group attribute can hide compound disadvantage. Under pipeline B, reported recall is 0.83 for old, low-citation English papers but 0.03 for old, low-citation non-English papers and 0.02 for recent, low-citation non-English papers. The worst cell is 0.016 for B, compared with 0.424 for A, 0.830 for D, 0.853 for E, and 0.624 for F. This supports reporting recall for language–recency–citation intersections instead of aggregate recall alone.

5.3 Sensitivity and confounding (RQ1 and RQ5)

Citation weight. The reported sweep shows aggregate recall and $\mathrm{PRR}_{\mathrm{age}}$ decreasing as citation weight β increases. Even $\beta = 0.1$ lowers the age-controlled ratio for English BM25 from about 0.79 to 0.67; at $\beta \geq 0.5$, recall for low-citation papers falls sharply. The numerical scale of β depends on score normalisation and should not be transferred directly to a real system.

RQ5: A low-citation gap can arise without citation-aware ranking. Pipeline A never uses citation counts, yet its $\mathrm{PRR}_{\mathrm{age}}$ is approximately 0.79. When the language-visibility penalty is zero ($\delta = 0$), the reported ratio is 0.958; as the penalty grows, it falls to 0.725. As the citation–quality correlation ρ increases from 0 to 0.9, the ratio falls from 0.815 to 0.723 because lower-purity papers are harder to retrieve from text. Multilingual pipeline D is less sensitive to the language penalty, with its reported ratio changing from 0.948 to 0.922. Therefore, a raw ratio below one is not proof of direct popularity bias; language, age, and text quality need to be controlled.

Cross-lingual overlap. With no language-independent tokens, pipeline A's LRR is 0; with a 25% shared-token share, it reaches 0.596. Pipeline D's LRR stays between 0.900 and 0.931 across the reported sweep. This demonstrates that baseline language disparities depend strongly on how much vocabulary the languages share in the simulator.

Table 4: Reported sensitivity to language-independent token share (mean of 10 corpora).

Shared-token share	A aggregate	A LRR	B LRR	D aggregate	D LRR
0.00	0.727	0.000	0.000	0.901	0.900
0.06	0.796	0.298	0.089	0.906	0.908
0.12 (default)	0.838	0.481	0.121	0.909	0.916
0.25	0.862	0.596	0.200	0.912	0.931

5.4 Feedback-loop simulation (RQ6)

The feedback simulation runs for 12 rounds. In each round, 600 readers query the English pipeline, inspect the top 20 results, and cite a retrieved paper with probability 0.25 if it is relevant and 0.01 otherwise. At the default citation weight $\beta = 0.3$, recall for initially low-cited papers changes only from 0.465 to 0.455, while their relative top-20 exposure remains at 0.295. The strongest setting ($\beta = 0.6$) shows a larger change: citation-count Gini rises from 0.605 to 0.649 and aggregate recall falls from 0.322 to 0.294, although low-citation recall is already near zero. At $\beta = 0$, Gini initially falls from 0.605 to about 0.575 as citations are redistributed through exposure to relevant papers. Thus RQ6 is not supported at the default setting; the simulator suggests a persistent initial disadvantage rather than strong additional amplification. This conclusion is limited by the simple reader model, fixed attention depth, rank-independent citation probability, and 12-round horizon.

5.5 Summary of research-question outcomes

Table 5: Reported outcomes; all verdicts are specific to the tested simulator.

RQ	Reported result	Verdict
RQ1	Citation boost lowers aggregate recall by 0.132 and PRR_{age} by 0.290 (A to B).	Supported
RQ2	English-only LRR is 0.476; it reaches zero with no shared tokens.	Supported
RQ3	Multilingual LRR is 0.919; translation quality matters and low accuracy can hurt.	Supported with conditions
RQ4	Diversity re-ranking raises LRR by 0.034 with an aggregate decrease of 0.005; over-correction appears at $p = 1$.	Partially supported
RQ5	A citation-neutral ranker's PRR_{age} falls from 0.958 to 0.725 as the language–citation gap grows.	Confounding supported
RQ6	Recall change over 12 rounds is at most 0.010 at $\beta = 0.3$.	Not supported

6 Discussion and Implications

The experiment supports three cautious conclusions within the synthetic testbed. First, English-only retrieval can miss many non-English relevant papers when the vocabularies overlap little. Second, multilingual query expansion can recover much of that loss when translations are sufficiently accurate. Third, adding citation counts directly to the ranking score can lower recall when citations are correlated with language and age but add no relevance signal beyond text.

The results do not establish real-world bias magnitudes. In actual scholarly search, citations can signal foundational work or relevance that is not obvious from document text. The simulator deliberately excludes that possible benefit. Similarly, the diversity re-ranker is designed around the same groups used for evaluation, so part of its measured parity gain is built into the intervention. A real evaluation should compare against controls, including a random re-ranking baseline and a diversity method that does not directly use the evaluation labels.

Practical implications are to (i) expose citation counts as metadata rather than silently blending them into relevance; (ii) test multilingual components over a held-out range of translation accuracy; and (iii) report recall by language, citation level, age, and their intersections, not only in aggregate.

7 Threats to Validity and Limitations

- **Synthetic data.** Results are not estimates for existing scholarly search systems. A real annotated multilingual benchmark is needed.
- **Simplified language model.** The corpus uses disjoint vocabularies with controllable overlap and noisy one-to-one translation. Real languages differ in morphology, script, and resource availability.
- **Weak semantic baseline.** LSA is a lightweight proxy and does not represent modern multilingual neural retrievers.
- **Limited metrics.** Recall@100 is the primary measure; nDCG, MAP, and precision are not reported. Other metrics may change the comparison.
- **Parameter dependence.** The study varies β, p, ρ, δ, and shared-token share, but not corpus size, topic overlap, or query length. The scales of β and λ depend on score normalisation.

- **Re-ranker alignment.** Pipeline E uses language, recency, and citation cells that also define the evaluation groups; its parity gain is partly built into the design.
- **Simple feedback model.** The simulation uses fixed attention depth, no repeated users, rank-independent citation probability, and only 12 rounds.
- **Statistical interpretation.** Confidence intervals describe variation across simulated corpora, not uncertainty about real-world systems.

8 Development Process and Use of AI Tools

AI-use disclosure. Claude was used to draft initial versions of Python modules, and ChatGPT was used for language editing. The author is responsible for the research question, experimental design, interpretation, and final manuscript. AI-generated code and prose were treated as assistance, not as evidence that the experiment is correct.

Verification status. The supplied paper describes scripts that generate tables and figures from saved JSON outputs, but the experiments have not been independently rerun or reconciled against every reported value in this revision. Before submission, run the main and sweep scripts, compare outputs against all tables and figures, and test metric implementations on hand-computed boundary cases. The findings should not be described as independently reproduced until those checks are complete.

9 Reproducibility

The described implementation uses Python with NumPy, SciPy, scikit-learn, and Matplotlib. The original paper reports versions NumPy 2.4.4, scikit-learn 1.8.0, and Matplotlib 3.10.8, and approximate single-core runtimes of 26 seconds for the main experiment, 116 seconds for sweeps, and 10 seconds for the feedback simulation. The commands reported in the supplied document are:

```
cd code
python run_main.py
python run_sweeps.py
python run_feedback.py
python make_figures.py
python analyze.py
```

These commands assume the project files and the ‘code/‘, ‘results/‘, and ‘figures/‘ directories are available. They were not executed during this LaTeX reconstruction.

10 Conclusion

This experiment converts claims from a prior literature review into measurable questions in a controlled synthetic testbed. English-only querying leaves many non-English papers unretrieved, citation boosting reduces low-citation recall and aggregate recall in this simulator, and multilingual citation-neutral retrieval with diversity-aware re-ranking yields the strongest reported balance (aggregate recall 0.904, LRR 0.953). Poor translation can be worse than no expansion, while the default feedback loop adds little beyond the initial ranking effect. The apparent low-citation gap of a citation-neutral ranker is substantially confounded by language and text quality. Because the data are synthetic, the contribution is a measurement protocol

and mechanism-level evidence rather than a claim about real-world bias magnitudes. Next steps are to verify every reported value against the saved outputs, add controls for the group-aware re-ranker, evaluate further retrieval metrics, and test the framework on a real annotated multilingual AI-literature corpus.

A Core Code

The following excerpts are transcribed from the supplied paper and have only been formatted for readability. They have not been independently executed or validated.

```
def group_recall(ranks, qtopic, topic, g, ng, k, nrep):
    top = ranks[:, :k]
    hit = topic[top] == qtopic[:, None]
    tp = np.bincount(g[top][hit], minlength=ng)
    return tp / (np.bincount(g, minlength=ng) * nrep)

def blend(R, cit, beta):
    c = np.log1p(cit)
    c = c / c.max()
    return (1 - beta) * R + beta * c[None, :]

def diversity_rerank(S, cell, kmax=200, pool=600, lam=0.25):
    P = np.argpartition(-S, pool, axis=1)[:, :pool]
    out = np.empty((S.shape[0], kmax), int)
    for q in range(S.shape[0]):
        cand, sc, cc = P[q], S[q, P[q]], cell[P[q]]
        cnt = np.zeros(cell.max() + 1)
        avail = np.ones(pool, bool)
        for s in range(kmax):
            val = np.where(avail, sc - lam * cnt[cc] / (s + 1), -np.inf)
            j = int(np.argmax(val))
            out[q, s] = cand[j]
            avail[j] = False
            cnt[cc[j]] += 1
    return out
```

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